DATA PRODUCT · 2025 SEASON · SYNTHETIC DATA

FLAG FOOTBALL: A DATA PRODUCT, END TO END

SYNTHETIC DATA

Three business goals, turned into five data products that are owned, measured and safe for kids.

5	2/6	~1,300	0
PRODUCTS	GATES MET	NEW FANS	BOUNDARY BREAKS

1 · WHAT WOULD YOU BUILD FOR THESE GOALS?

EACH GOAL HAS ONE PRODUCT THAT MOVES IT DIRECTLY.

Three goals led to five products.

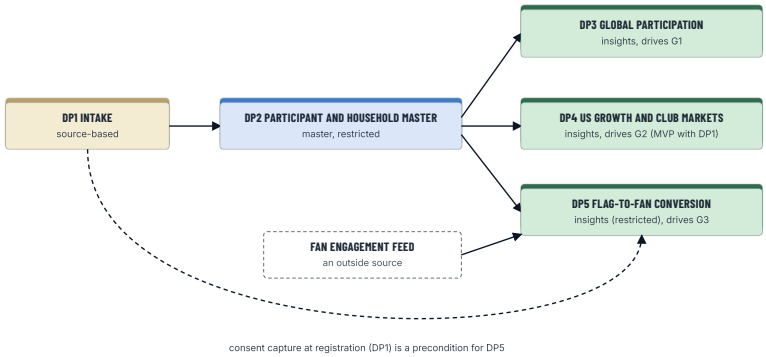
GOAL	DP1 INTAKE	DP2 MASTER	DP3 GLOBAL	DP4 US GROWTH	DP5 FAN CONVERSION
G1 Grow the NFL globally	UTILITY	ENABLER	DRIVER		
G2 Grow flag in the US as a safe, low-cost way into football	UTILITY	ENABLER		DRIVER	
G3 Turn families into fans		ENABLER			DRIVER

Driver: moves the goal directly. Enabler: makes other products possible. Utility: keeps things running.

2 · WHAT WOULD YOU BUILD FOR THESE GOALS? (CONTINUED)

INTAKE FEEDS THE MASTER, AND THE MASTER FEEDS THE THREE INSIGHT PRODUCTS.

Conversion also needs the fan feed and guardian consent.



3 · WHO ARE THE PRODUCTS FOR?

SIX PERSONAS, AND WHAT WE'D ASK THEM.

There are no real users yet, so these are assumptions to test.

GL INTERNATIONAL GROWTH LEAD
SERVES G1

Decide which countries to invest in

We would ask: Which decisions did you make last year without data?

OL NFL FLAG DOMESTIC OPERATIONS LEAD
SERVES G2

Know where flag is growing and where leagues are struggling

We would ask: Which numbers do you distrust today, and why?

CM CLUB YOUTH AND COMMUNITY MANAGER
SERVES G2, G3

Grow flag in our market and connect families to the club

We would ask: What would you do differently with a market view?

MA FAN AND CLUB MARKETING ANALYST
SERVES G3

Show that flag creates fans, and reach families who agreed to be reached

We would ask: What evidence would convince finance that flag drives fandom?

CL PRIVACY AND COMPLIANCE LEAD
SERVES G3 (GUARDRAIL)

Make sure children's data is never misused

We would ask: What would make you comfortable approving club access?

LO LEAGUE OPERATOR (VOLUNTEER)
SERVES G2 (SUPPLIER)

Run the league and send the data that is asked for

We would ask: What is the most you would fill in at registration?

4 · WHAT IS EACH PRODUCT?

EACH PRODUCT HAS A LABEL: OWNER, VERSION, LIVE QUALITY GRADE AND RULES FOR USE.

Open one to see who uses it and what it must never do.

PRODUCT	TYPE	STATUS	VERSION	QUALITY	OWNER	RULE
DP1 Intake	Source-based	MVP · building	v2.0.0	Grade C	Data Product Manager, NFL FLAG Insights (fictional)	Never: re-identification of a child from an intake file
DP2 Master	Master	Proposed	v0.10.0	Identity error under 1%	Data Product Manager, NFL FLAG Insights (fictional)	Never: marketing access to the master
DP3 Global	Insights	Proposed	v0.9.0	Grade C	Data Product Manager, NFL FLAG Insights (fictional)	Never: publishing a country count below the small-cell size
DP4 US growth	Insights	MVP · building	v1.1.0	Grades vary by market	Data Product Manager, NFL FLAG Insights (fictional)	Never: individual targeting
DP5 Fan conversion	Insights	Proposed	v0.1.0	Grade D	Data Product Manager, NFL FLAG Insights (fictional)	Never: child-level targeting

Quality grade: A is 95 or more, B is 85, C is 70, D is below 70 (score out of 100). Identity error is how far identity matching alone could move the count. Grades vary by market: each market is graded on its own.

5 · WHAT DO WE BUILD FIRST, AND WHY?

INTAKE AND US GROWTH FIRST. CONVERSION WAITS: HIGH VALUE, DATA NOT READY.

Each product is scored on value, feasibility, readiness and privacy risk.

PRODUCT	VALUE	FEASIBILITY	READINESS	PRIVACY RISK	TOTAL	READING
DP1 Intake	4	5	3	1	**11**	Build first
DP4 US growth	5	4	3	2	**10**	Build first
DP2 Master	4	4	4	4	**8**	Next
DP3 Global	4	3	2	2	**7**	High value, low readiness: sequenced, not deprioritized
DP5 Fan conversion	5	2	1	5	**3**	High value, low readiness: sequenced, not deprioritized

Scores 1 to 5. Total = value + feasibility + readiness - privacy risk.

6 · WHAT HAS TO BE TRUE BEFORE EACH PRODUCT MOVES?

2 OF 6 RELEASE GATES MET. THE MVP WAITS ON THE US DATA SCORE (82 → 85).

No dates: a product moves when its gate is met and the right person signs off.

	DELIVERED	NOW	NEXT	LATER
SOURCE-BASED **DP1 League Registration Intake** Grow NFL presence globally, Grow flag in the US	**Raw + staging and quality profile** (v0.1 · Phase 1) **Contracts + onboarding playbook** (v0.5 · Phase 4)	**Data standard, league fix lists (by records affected), quality port** (v1.0) ○ Not yet now 82.1 points, needs 85 points or more ↩ from gap 1 ↩ from gap 4		**"Played youth flag?" question at adult sign-up** (v1.2) ↩ from gap 5
MASTER **DP2 Flag Participant & Household Master** Grow NFL presence globally, Grow flag in the US, Track conversion to fans, protecting children	**Identity resolution + review queue** (v1.0 · Phase 2)		**Restricted zones; consumers use product ports only. One identity chain: merge the Phase 5 feeds into Phase 2 resolution (known debt)** (v1.1) Met ✓ now 0.43%, limit 1% ↩ from gap 3	
INSIGHTS **DP3 Global Participation Insights** Grow NFL presence globally	**Counting method, scorecard, ranges** (v0.9 · Phases 3 + 5)		**Country view for growth decisions** (v1.0) ○ Not yet now 1, needs 6 or more countries	
INSIGHTS **DP4 US Flag Growth & Club Market Insights** Grow flag in the US	**What we learned.** We learned the real pain point was thin, poor-quality data, not consent to contact. Intake and data quality moved first. **Retention + club market prototype** (v0.5 · Phases 4 + 5)	**MVP: market view for clubs and the fix ranking by participants** (v1.0) ○ Not yet now 82.1 points, needs 85 points or more ◆ **Privacy lead:** Signs off the club share		**Girls' share by market** (v1.1) ◆ **Clubs:** Confirm market boundaries (ADR-016 approximation)
INSIGHTS, RESTRICTED **DP5 Flag-to-Fan Conversion Insights** Track conversion to fans, protecting children	**Fan-lift analysis prototype (restricted zone)** (v0.1 · Phase 4)		**Fans created by flag (aggregates only)** (v0.5) Met ✓ now 80.2%, needs 80% or more ◆ **Privacy lead:** Approves the analytics linkage (ADR-009)	**Families clubs may contact** (v1.0) ○ Not yet now 28.5%, needs 75% or more ◆ **Product + legal:** Add a separate guardian club-contact consent at registration ↩ from gap 2

NOT DOING ~~marketing to anyone under 18~~ ~~fuzzy matching of families to fan accounts~~ ~~buying outside data about kids~~ ~~scoring the value of flag families~~ ~~carrying a youth record past 18~~

No dates: products move when their data is good enough and the right person has signed off. Bars show where a gate stands; the tick is the target. ◆ marks a decision. Targets are set for a synthetic dataset.

7 · HOW DO WE PROTECT KIDS?

MINORS ARE GUARDIAN-ONLY. WE COUNT FAMILIES, NEVER KIDS.

9% of families with a child under 13 agreed to club contact.

UNDER 13	13–15	16–17	18+
Guardian	Guardian	Guardian	Self, fresh consent
Counted in totals · club contact only via guardian consent	Counted in totals · club contact only via guardian consent	Counted in totals · club contact only via guardian consent	Own fan record, only with fresh consent
9% agreed to club contact: 520 of 5,790 families	**9%** agreed to club contact: 190 of 2,230 families	**8%** agreed to club contact: 190 of 2,250 families	**5%** agreed to club contact: 60 of 1,350 families

~~Marketing to under-18s~~ · ~~Buying data about kids~~ · ~~Fuzzy family matching~~ · ~~Scoring families~~ · ~~Youth records past 18~~

8 · IS IT WORKING?

FANS CREATED BY FLAG: ABOUT 1,300 (900–1,600).

One row per goal, then the health of each product.

G1

7,400

PLAYERS WORLDWIDE

7,400 players in 8 countries. Only 1 reports at grade B or better.

Not yet: now 1, needs 6 or more countries

So what. We can count globally, but can't compare countries fairly yet.

G2

47%

GIRLS' SHARE OF YOUTH

Girls 47% · 58% play again · 14 of 76 leagues sent no file.

Not yet: now 82.1 points, needs 85 points or more

So what. Growth looks real, but missing leagues mean we undercount it.

G3

1,300

NEW FANS FROM FLAG

About 1,300 new fans (900–1,600) · engaged families +20% · 36% of families linked.

Not yet: now 28.5%, needs 75% or more

So what. Flag does create fans. With only 36% of families linked, this is a floor, not the full number.

2025, synthetic

PRODUCT	QUALITY	TESTS	PRIVACY CHECKS	STATUS
DP1 Intake	Grade C	✓	✓	WATCH Quality: C
DP2 Master	Identity error under 1%	✓	✓	WATCH Age-rule rows still unverified: 23
DP3 Global	Grade C	✓	✓	WATCH Quality: C
DP4 US growth	Grades vary by market	✓	✓	WATCH Quality: A: 12, B: 4, C: 5, D: 5
DP5 Fan conversion	Grade D	✓	✓	WATCH Quality: D

9 · WHERE ARE THE GAPS, AND WHAT DID THEY CHANGE?

7 GAPS FOUND. 5 OF THEM CHANGED THE ROADMAP.

Each gap shows its evidence and the card it moved.

GAP	EVIDENCE	PRODUCT	WHAT IT CHANGED	ROADMAP
1 **US data score is below target**	now 82.1 points, needs 85 points or more	DP4	Blocks the DP1 and DP4 release; intake and quality went first (the pivot)	CHANGED ROADMAP
2 **Consent fully known for too few families**	now 28.5%, needs 75% or more	DP5	Club contact moved to Later; guardian consent added at sign-up as a decision	CHANGED ROADMAP
3 **A second identity chain was found**	planned for v1.1	DP2	Added the "one identity chain" card	CHANGED ROADMAP
4 **Leagues that sent no file**	14 of 76 leagues	DP1	The send-your-file fix ranks first on every fix list	CHANGED ROADMAP
5 **Youth-to-adult link cannot be measured**	Fresh consent at 18 and no carry-over: a youth record is not linked to the adult by design	DP5	Added the played-youth-flag question at adult sign-up	CHANGED ROADMAP
6 **Age rules not yet verified**	23 of 28 rules	DP1	No roadmap change; the strictest rule applies	WATCH
7 **Too few countries at grade B**	now 1, needs 6 or more countries	DP3	No roadmap change; it is the gate for the country view	WATCH

10 · WHAT DID THE DATA TEACH US?

FIVE THINGS WE ONLY LEARNED BY BUILDING IT.

We started with consent to contact. The real problem was thin, messy data.

FINDING	EVIDENCE	SO WHAT
Flag mostly creates new fans	~1,300 new fan accounts; existing fan families +20%	Measure fans created separately from fans engaged. Flag is an acquisition channel.
Missing data isn't random	Gender is blank more often in younger mixed divisions; girls' share 46.5–47.1%	Never publish girls' share without its range.
Some growth is better reporting	14 of 76 leagues sent no file	Report league coverage next to every growth number.
Good enough depends on the job	Counting OK; contacting families blocked (score 82/85; consent fully known 28.5%)	Grade data per use, not with one score.
Contact is limited by consent, not data	9% of families with young kids said yes	Fix it with a clearer question at registration, not data cleaning.

NEXT DECISION Privacy lead approves the fan analysis; product and legal add a clear guardian consent question at sign-up. It unlocks clubs reaching families, with consent.

Independent portfolio project using synthetic data. Not affiliated with or endorsed by any league or team.

All figures synthetic, built to test the method · Methodology 2.0 · Full detail: Data product tab →